INCH-POUND
MIL-PRF-3098/56H
7 January 2010
SUPERSEDING
MIL-PRF-3098/56G
12 January 2005

PERFORMANCE SPECIFICATION SHEET

# CRYSTAL UNIT, QUARTZ, CR85/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 0.8 MHz to 20 MHz; fundamental; noncontrolled; series resonance.

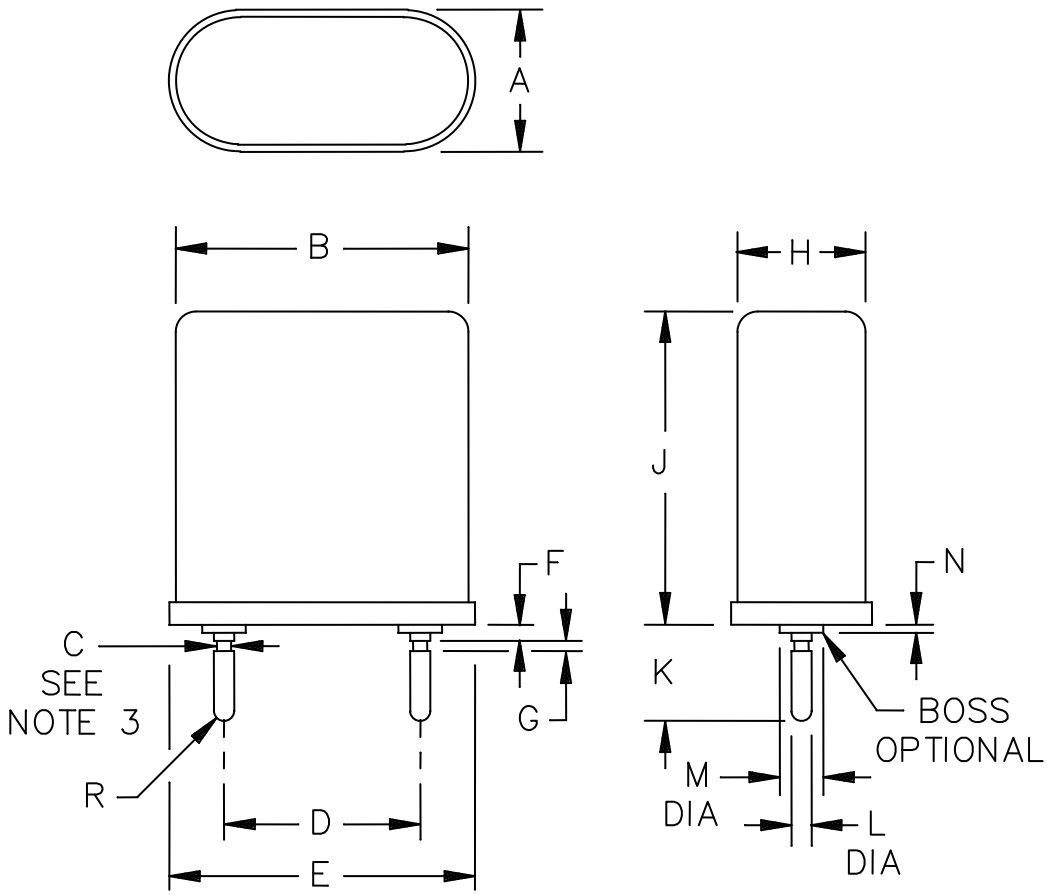


FIGURE 1. Crystal unit - CR85/U.



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| Ltr | Inches | | mm | |
|---|---|---|---|---|
| | Min | Max | Min | Max |
| A | --- | .352 | --- | 8.94 |
| B | --- | .725 | --- | 18.41 |
| C | .030 | .037 | 0.76 | 0.94 |
| D | .478 | .494 | 12.14 | 12.55 |
| E | --- | .757 | --- | 19.23 |
| F | .030 | .040 | 0.76 | 1.02 |
| G | .015 | .025 | 0.38 | 0.64 |
| H | --- | .317 | --- | 8.05 |
| J | --- | .775 | --- | 19.69 |
| K | .223 | .248 | 5.66 | 6.30 |
| L | .048 | .052 | 1.22 | 1.32 |
| M | .075 | .141 | 1.91 | 3.58 |
| N | .015 | .025 | 0.38 | 0.63 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR85/U - Continued.

REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 0.8 MHz to 20 MHz, inclusive.

Capacitance, shunt: 7.0 pF, maximum.

Frequency tolerances:

Primary operating temperature range: ±20 parts per million (ppm).

Secondary operating temperature range: ±30 ppm.

Equivalent resistance: See table I.



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TABLE I. Equivalent resistance.

| Frequency range, inclusive | Maximum resistance |
|---|---|
| MHz | Ohms |
| 0.80 to 0.85 | 520 |
| 0.85+ to 0.90 | 480 |
| 0.90+ to 1.00 | 440 |
| 1.00+ to 1.12 | 400 |
| 1.12+ to 1.25 | 380 |
| 1.25+ to 1.37 | 340 |
| 1.37+ to 1.50 | 300 |
| 1.50+ to 1.62 | 280 |
| 1.62+ to 1.75 | 250 |
| 1.75+ to 1.87 | 220 |
| 1.87+ to 2.00 | 190 |
| 2.00+ to 2.12 | 170 |
| 2.12+ to 2.25 | 150 |
| 2.25+ to 2.60 | 130 |
| 2.60+ to 3.00 | 90 |
| 3.00+ to 3.40 | 70 |
| 3.40+ to 3.75 | 52 |
| 3.75+ to 4.00 | 45 |
| 4.00+ to 5.00 | 37 |
| 5.00+ to 7.00 | 25 |
| 7.00+ to 10.00 | 20 |
| 10.00+ to 15.00 | 18 |
| 15.00+ to 20.00 | 15 |

Mode of oscillation: Fundamental.

Operating temperature range:

Primary operating temperature range: -40°C to +90°C, inclusive.

Secondary operating temperature range: -55°C to -40°C, and +90°C to +105°C, inclusive.

Drive level, rated: 1.0 mW maximum.

Resonance: Series.

Capacitance, shunt: 7.0 pF maximum.



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| Shock (specified pulse): | 0.8 MHz to 2.0 MHz | 2.0+ MHz to 20.0 MHz |
|---|---|---|
| Frequency change permitted: | ±10 ppm | ±5 ppm |
| Equivalent resistance change permitted: | ±15 percent | ±10 percent |

| Vibration: Method 201 of MIL-STD-202, | 0.8 MHz to 2.0 MHz | 2.0+ MHz to 20.0 MHz |
|---|---|---|
| Frequency change permitted: | ±10 ppm | ±5 ppm |
| Equivalent resistance change permitted: | ±15 percent | ±10 percent |

| Thermal shock: | 0.8 MHz to 2.0 MHz | 2.0+ MHz to 20.0 MHz |
|---|---|---|
| Frequency change permitted: | ±10 ppm | ±5 ppm |
| Equivalent resistance change permitted: | ±15 percent | ±10 percent |

Aging:

Frequency change permitted: ±5 ppm.

Part or Identifying Number (PIN): CR85/U (followed by specified frequency), see 1.2 of MIL-PRF-3098.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.



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Custodians:
Army - CR
Navy - EC
Air Force - 99
DLA - CC

Review activities:
Army - AR, MI
Navy - AS, MC, SH
Air Force - 19, 84

Preparing activity:
Army - CR

Agent:
DLA - CC

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